MLP / CIFAR-10 | gradient clipping before Adam: 1.0 versus 0.5
Each compared with unchanged LR | seed 0 | full 5,000-image objective | same Spark

Clip threshold 1.0 | independent panel ranges

Clip threshold 0.5 | independent panel ranges

Unchanged baseline
Global norm clipping C = 1.0
Global norm clipping C = 0.5
Training loss (log scale)
Raw gradient ρ_t
Baseline: raw
Clipping run: raw
Clipping run: after clipping
Gradient norm
$\eta\hat{\kappa}_t$
Applied LR / base LR
Both LR curves coincide at base LR 0.001
Completed optimizer updates

Shaded: shared prefix. Fork at512; exact replay verified, no hardware change. Solid curves, no smoothing.
Green: recorded post-clipping Adam input, states512-799. Rho and eta kappa hat remain raw.